

Module 3: Data Analysis Expressions (DAX)

DAX Formula List

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Aggregation Functions

1. APPROXIMATEDISTINCTCOUNT

Formula: `APPROXIMATEDISTINCTCOUNT (<columnName>)`

Returns an estimated count of unique values in a column. This function invokes a corresponding aggregation operation in the data source, which is optimized for query performance, but with slightly reduced accuracy.

2. AVERAGE

Formula: `AVERAGE (<column>)`

Returns the average (arithmetic mean) of all the numbers in a column.

3. COUNT

Formula: `COUNT (<column>)`

Counts the number of rows in the specified column that contain non-blank values.

4. DISTINCTCOUNT

Formula: `DISTINCTCOUNT (<column>)`

Counts the number of distinct values in a column.

5. MAX

Formula: `MAX (<column>)`

Returns the largest value in a column, or between two scalar expressions.

6. MIN

Formula: `MIN (<column>)`

Returns the smallest value in a column, or between two scalar expressions.

7. PRODUCT

Formula: `PRODUCT (<column>)`

Returns the product of the numbers in a column.

8. SUM

Formula: `SUM (<column>)`

Adds all the numbers in a column.

9. SUMX

Formula: `SUMX (<table>, <expression>)`

Returns the sum of an expression evaluated for each row in a table.

Date and Time Functions

10.CALENDAR

Formula: `CALENDAR(<start_date>, <end_date>)`

Returns a table with a single column named "Date" that contains a contiguous set of dates. The range of dates is from the specified start date to the specified end date, inclusive of those two dates.

11.CALENDARAUTO

Formula: `CALENDARAUTO([fiscal_year_end_month])`

Returns a table with a single column named "Date" that contains a contiguous set of dates. The range of dates is calculated automatically based on data in the model.

12.DATE

Formula: `DATE(<year>, <month>, <day>)`

Returns the specified date in datetime format.

13.DATEDIFF

Formula: `DATEDIFF(<Date1>, <Date2>, <Interval>)`

Returns the number of interval boundaries between two dates.

14.DATEVALUE

Formula: `DATEVALUE(date_text)`

Converts a date in text format to a date in datetime format.

15.DAY

Formula: `DAY(<date>)`

Returns the day of the month, a number from 1 to 31.

16.EDATE

Formula: `EDATE(<start_date>, <months>)`

Returns the date that is the indicated number of months before or after the start date. Use EDATE to calculate maturity dates or due dates that fall on the same day of the month as the date of issue.

17.EOMONTH

Formula: `EOMONTH(<start_date>, <months>)`

Returns the date in datetime format of the last day of the month, before or after a specified number of months. Use EOMONTH to calculate maturity dates or due dates that fall on the last day of the month.

18.NETWORKDAYS

Formula: NETWORKDAYS (<start_date>, <end_date>[, <weekend>, <holidays>])

Returns the number of whole workdays between two dates (inclusive).

Parameters specify which and how many days are weekend days. Weekend days and days specified as holidays are not considered as workdays.

19.NOW

Formula: NOW ()

The NOW function is useful when you need to display the current date and time on a worksheet or calculate a value based on the current date and time, and have that value updated each time you open the worksheet.

20.TIME

Formula: TIME (hour, minute, second)

Converts hours, minutes, and seconds given as numbers to a time in datetime format.

21.TODAY

Formula: TODAY ()

Returns the current date.

22.UTCNOW

Formula: UTCNOW ()

Returns the current UTC date and time.

23.YEAR

Formula: YEAR (<date>)

Returns the year of a date as a four digit integer in the range 1900-9999.

Filter functions

24.ALL

Formula: ALL([<table> | <column>[, <column>[, <column>[, ...]]]])

Returns all the rows in a table, or all the values in a column, ignoring any filters that might have been applied. This function is useful for clearing filters and creating calculations on all the rows in a table.

25. ALLCROSSFILTERED

Formula: `ALLCROSSFILTERED(<table>)`

Clear all filters which are applied to a table.

26. ALLEXCEPT

Formula: `ALLEXCEPT(<table>, <column> [, <column> [, ...]])`

Removes all context filters in the table except filters that have been applied to the specified columns.

27. ALLSELECTED

Formula: `ALLSELECTED([<tableName> | <columnName> [, <columnName> [, <columnName> [, ...]]]])`

Removes context filters from columns and rows in the current query, while retaining all other context filters or explicit filters.

The `ALLSELECTED` function gets the context that represents all rows and columns in the query, while keeping explicit filters and contexts other than row and column filters. This function can be used to obtain visual totals in queries.

28. CALCULATE

Formula: `CALCULATE(<expression> [, <filter1> [, <filter2> [, ...]])`

Similar to `CALCULATETABLE` it evaluates an expression in a modified filter context.

29. FILTER

Formula: `FILTER(<table>, <filter>)`

Returns a table that represents a subset of another table or expression.

30. INDEX

Formula: `INDEX(<position> [, <relation>] [, <orderBy>] [, <blanks>] [, <partitionBy>] [, <matchBy>])`

Returns a row at an absolute position, specified by the position parameter, within the specified partition, sorted by the specified order. If the current partition can't be deduced to a single partition, multiple rows may be returned.

31. MATCHBY

Formula: `MATCHBY ( [ <matchBy_columnName> [, matchBy_columnName [, ...]] ] )`

In window functions, defines the columns that are used to determine how to match data and identify the current row.

32.ORDERBY

Formula: ORDERBY ([`<orderBy_expression>`[, `<order>`[, `<orderBy_expression>`[, `<order>`]] ...]])

Defines the expressions that determine the sort order within each of a window function's partitions.

33.ROWNUMBER

Formula: ROWNUMBER ([`<relation>`][, `<orderBy>`][, `<blanks>`][, `<partitionBy>`][, `<matchBy>`])

Returns the unique ranking for the current context within the specified partition, sorted by the specified order. If a match cannot be found then then rownumber is blank.

Financial functions

34.ACCRINT

Formula: ACCRINT(`<issue>`, `<first_interest>`, `<settlement>`, `<rate>`, `<par>`, `<frequency>`[, `<basis>`[, `<calc_method>`]])

Returns the accrued interest for a security that pays periodic interest.

35.AMORDEGRC

Formula: AMORDEGRC(`<cost>`, `<date_purchased>`, `<first_period>`, `<salvage>`, `<period>`, `<rate>`[, `<basis>`])

Returns the depreciation for each accounting period. This function is provided for the French accounting system. If an asset is purchased in the middle of the accounting period, the prorated depreciation is taken into account. The function is similar to AMORLINC, except that a depreciation coefficient is applied in the calculation depending on the life of the assets.

36.AMORLINC

Formula: AMORLINC(`<cost>`, `<date_purchased>`, `<first_period>`, `<salvage>`, `<period>`, `<rate>`[, `<basis>`])

Returns the depreciation for each accounting period. This function is provided for the French accounting system. If an asset is purchased in the middle of the accounting period, the prorated depreciation is taken into account.

37.COUPDAYBS

Formula: COUPDAYBS(`<settlement>`, `<maturity>`, `<frequency>`[, `<basis>`])

Returns the number of days from the beginning of a coupon period until its settlement date.

38.CUMIPMT

Formula: CUMIPMT(<rate>, <nper>, <pv>, <start_period>, <end_period>, <type>)

Returns the cumulative interest paid on a loan between start_period and end_period.

39.DB

Formula: DB(<cost>, <salvage>, <life>, <period>[, <month>])

Returns the depreciation of an asset for a specified period using the fixed-declining balance method.

40.DISC

Formula: DISC(<settlement>, <maturity>, <pr>, <redemption>[, <basis>])

Returns the discount rate for a security.

41.DURATION

Formula: DURATION(<settlement>, <maturity>, <coupon>, <yld>, <frequency>[, <basis>])

Returns the Macauley duration for an assumed par value of \$100. Duration is defined as the weighted average of the present value of cash flows, and is used as a measure of a bond price's response to changes in yield.

42.NOMINAL

Formula: NOMINAL(<effect_rate>, <npery>)

Returns the nominal annual interest rate, given the effective rate and the number of compounding periods per year.

43.PRICE

Formula: PRICE(<settlement>, <maturity>, <rate>, <yld>, <redemption>, <frequency>[, <basis>])

Returns the price per \$100 face value of a security that pays periodic interest.

44.VDB

Formula: VDB(<cost>, <salvage>, <life>, <start_period>, <end_period>[, <factor>[, <no_switch>]])

Returns the depreciation of an asset for any period you specify, including partial periods, using the double-declining balance method or some other method you specify. VDB stands for variable declining balance.

45. YIELD

Formula: `YIELD(<settlement>, <maturity>, <rate>, <pr>, <redemption>, <frequency>[, <basis>])`

Returns the yield on a security that pays periodic interest. Use YIELD to calculate bond yield.

Information functions

46. COLUMNSTATISTICS

Formula: `COLUMNSTATISTICS ()`

Returns a table of statistics regarding every column in every table in the model.

47. CONTAINSROW

Formula: `CONTAINSROW(<Table>, <Value> [, <Value> [, ...] ] )`

Returns TRUE if there exists at least one row where all columns have specified values.

48. CONTAINSSTRINGEXACT

Formula: `CONTAINSSTRINGEXACT(<within_text>, <find_text>)`

Returns TRUE or FALSE indicating whether one string contains another string.

49. CUSTOMDATA

Formula: `CUSTOMDATA ()`

Returns the content of the CustomData property in the connection string.

50. HASONEVALUE

Formula: `HASONEVALUE (<columnName>)`

Returns TRUE when the context for columnName has been filtered down to one distinct value only. Otherwise is FALSE.

51. ISBLANK

Formula: `ISBLANK (<value>)`

Checks whether a value is blank and returns TRUE or FALSE.

52. ISCROSSFILTERED

Formula: `ISCROSSFILTERED (<TableNameOrColumnName>)`

Returns TRUE when the specified table or column is cross-filtered.

53.ISERROR

Formula: ISERROR (<value>)

Checks whether a value is an error, and returns TRUE or FALSE.

54.ISLOGICAL

Formula: ISLOGICAL (<value>)

Checks whether a value is a logical value, (TRUE or FALSE), and returns TRUE or FALSE.

55.ISNUMBER

Formula: ISNUMBER (<value>)

Checks whether a value is a number and returns TRUE or FALSE.

56.ISODD

Formula: ISODD (number)

Returns TRUE if number is odd, or FALSE if number is even.

57.ISSELECTEDMEASURE

Formula: ISSELECTEDMEASURE (M1, M2, ...)

Used by expressions for calculation items to determine the measure that is in context is one of those specified in a list of measures.

58.ISSELECTEDMEASURE

Formula: ISSELECTEDMEASURE (M1, M2, ...)

Used by expressions for calculation items to determine the measure that is in context is one of those specified in a list of measures.

59.ISSUBTOTAL

Formula: ISSUBTOTAL (<columnName>)

Creates another column in a SUMMARIZE expression that returns True if the row contains subtotal values for the column given as argument, otherwise returns False.

60.SELECTEDMEASURE

Formula: SELECTEDMEASURE ()

Used by expressions for calculation items to reference the measure that is in context.

61.USERPRINCIPALNAME

Formula: USERPRINCIPALNAME ()

Returns the user principal name.

Logical functions

62.AND

Formula: `AND(<logical1>, <logical2>)`

Checks whether both arguments are TRUE, and returns TRUE if both arguments are TRUE. Otherwise returns false.

63.BITOR

Formula: `BITOR(<number>, <number>)`

Returns a bitwise OR of two numbers.

64.COALESCE

Formula: `COALESCE(<expression>, <expression>[, <expression>]...)`

Returns the first expression that does not evaluate to BLANK. If all expressions evaluate to BLANK, BLANK is returned.

65.IF

Formula: `IF(<logical_test>, <value_if_true>[, <value_if_false>])`

Checks a condition, and returns one value when it's TRUE, otherwise it returns a second value.

66.NOT

Formula: `NOT(<logical>)`

Changes FALSE to TRUE, or TRUE to FALSE.

Math and Trig functions

67. ABS

Formula: `ABS(<number>)`

Returns the absolute value of a number.

68.ACOS

Formula: `ACOS(number)`

Returns the arccosine, or inverse cosine, of a number. The arccosine is the angle whose cosine is number. The returned angle is given in radians in the range 0 (zero) to pi.

69.ACOT

Formula: `ACOT (number)`

Returns the principal value of the arc-cotangent, or inverse cotangent of a number.

70.ASIN

Formula: `ASIN (number)`

Returns the arcsine, or inverse sine, of a number. The arcsine is the angle whose sine is number. The returned angle is given in radians in the range $-\pi/2$ to $\pi/2$.

71.ATAN

Formula: `ATAN (number)`

Returns the arctangent, or inverse tangent, of a number. The arctangent is the angle whose tangent is number. The returned angle is given in radians in the range $-\pi/2$ to $\pi/2$.

72.CEILING

Formula: `CEILING (<number>, <significance>)`

Rounds a number up, to the nearest integer or to the nearest multiple of significance.

73.CONVERT

Formula: `CONVERT (<Expression>, <Datatype>)`

Converts an expression of one data type to another.

74.COS

Formula: `COS (number)`

Returns the cosine of the given angle.

75.CURRENCY

Formula: `CURRENCY (<value>)`

Evaluates the argument and returns the result as currency data type.

76.EVEN

Formula: `EVEN (number)`

Returns number rounded up to the nearest even integer. You can use this function for processing items that come in twos. For example, a packing crate accepts rows of one or two items. The crate is full when the number of items, rounded up to the nearest two, matches the crate's capacity.

77.FACT

Formula: `FACT (<number>)`

Returns the factorial of a number, equal to the series $1*2*3*...*$, ending in the given number.

78.FLOOR

Formula: `FLOOR (<number>, <significance>)`

Rounds a number down, toward zero, to the nearest multiple of significance.

79.LOG

Formula: `LOG (<number>, <base>)`

Returns the logarithm of a number to the base you specify.

80.MOD

Formula: `MOD (<number>, <divisor>)`

Returns the remainder after a number is divided by a divisor. The result always has the same sign as the divisor.

81.POWER

Formula: `POWER (<number>, <power>)`

Returns the result of a number raised to a power.

82.ROUND

Formula: `ROUND (<number>, <num_digits>)`

Rounds a number to the specified number of digits.

Other functions

83.TOCSV

Formula: `TOCSV (<Table>, [MaxRows], [Delimiter], [IncludeHeaders])`

Returns a table as a string in CSV format.

84.TOJSON

Formula: `TOJSON (<Table>, [MaxRows])`

Relationship functions

85.CROSSFILTER

Formula: `CROSSFILTER(<columnName1>, <columnName2>, <direction>)`

Specifies the cross-filtering direction to be used in a calculation for a relationship that exists between two columns.

86.USERELATIONSHIP

Formula: `USERELATIONSHIP(<columnName1>, <columnName2>)`

Specifies the relationship to be used in a specific calculation as the one that exists between columnName1 and columnName2.

Table manipulation functions

87.ADDCOLUMNS

Formula: `ADDCOLUMNS(<table>, <name>, <expression>[, <name>, <expression>]...)`

Adds calculated columns to the given table or table expression.

88.CROSSJOIN

Formula: `CROSSJOIN(<table>, <table>[, <table>]...)`

Returns a table that contains the Cartesian product of all rows from all tables in the arguments. The columns in the new table are all the columns in all the argument tables.

89.DATATABLE

Formula: `DATATABLE (ColumnName1, DataType1, ColumnName2, DataType2..., {{Value1, Value2...}, {ValueN, ValueN+1...}}...)`

Provides a mechanism for declaring an inline set of data values.

90.DISTINCT (column)

Formula: `DISTINCT(<column>)`

Returns a one-column table that contains the distinct values from the specified column. In other words, duplicate values are removed and only unique values are returned.

91.FILTERS

Formula: `FILTERS(<columnName>)`

Returns the values that are directly applied as filters to columnName.

92.GROUPBY

Formula: GROUPBY (<table> [, <groupBy_columnName> [, <groupBy_columnName> [, ...]] [, <name>, <expression> [, <name>, <expression> [, ...]])

The GROUPBY function is similar to the SUMMARIZE function. However, GROUPBY does not do an implicit CALCULATE for any extension columns that it adds. GROUPBY permits a new function, CURRENTGROUP, to be used inside aggregation functions in the extension columns that it adds. GROUPBY is used to perform multiple aggregations in a single table scan.

93.INTERSECT

Formula: INTERSECT (<table_expression1>, <table_expression2>)

Returns the row intersection of two tables, retaining duplicates.

94.NATURALINNERJOIN

Formula: NATURALINNERJOIN (<LeftTable>, <RightTable>)

Performs an inner join of a table with another table.

95.NATURALLEFTOUTERJOIN

Formula: NATURALLEFTOUTERJOIN (<LeftTable>, <RightTable>)

Performs a join of the Left Table with the Right Table by using the Left Outer Join semantics.

96.SUMMARIZE

Formula: SUMMARIZE (<table>, <groupBy_columnName>[, <groupBy_columnName>]...[, <name>, <expression>]...)

Returns a summary table for the requested totals over a set of groups.

97.TOPN

Formula: TOPN (<N_Value>, <Table>, <OrderBy_Expression>, [<Order>[, <OrderBy_Expression>, [<Order>]]...)

Returns the top N rows of the specified table.

98.UNION

Formula: UNION (<table_expression1>, <table_expression2> [, <table_expression>]...)

Creates a union (join) table from a pair of tables.

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